

Autosomal dominant cerebellar ataxia deafness and narcolepsy

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Abstract

A new autosomal dominant syndrome in a Swedish pedigree is described. Five patients were affected with cerebellar ataxia and sensorineural deafness. Four of these patients had symptoms of narcolepsy. Optic atrophy, other neurological abnormalities and psychiatric symptoms developed with increasing disease duration. Three patients had non-neurological disease in addition, including diabetes mellitus in two and hypertrophic cardiomyopathy in one. Autopsy with neuropathological examination was performed in one case. Molecular studies focused on the short arm of chromosome 6, including the HLA DR2 locus associated with narcolepsy and the (CAG)_n repeat at the spinocerebellar ataxia type 1 (SCA1) locus. Biochemical investigation of muscle biopsy of one case indicated mitochondrial dysfunction with selective decrease in ATP production for substrates that normally give the highest rates. The activity of glutamate dehydrogenase was reduced, indicating a low mitochondrial density. We postulate an autosomal dominant genetic factor responsible for this syndrome. Linkage was excluded to HLA DR2, and a normal sized SCA1 repeat was observed. We conclude that a locus predisposing to ataxia, deafness and narcolepsy exists outside this region of chromosome 6.

Keywords: Hereditary syndrome; Cerebellar ataxia; Sensorineural deafness; Narcolepsy; Diabetes mellitus; Epilepsy; Cardiomyopathy; Mitochondrial dysfunction; HLA system

1. Introduction

Cerebellar ataxic disorders can be divided into those with onset before the age of 20 years, which are nearly all of autosomal recessive inheritance and those with later onset, which are usually autosomal dominant (Harding, 1983, 1993). A few cerebellar ataxias are X-linked (Lutz et al., 1989). Ataxia also sometimes occurs in maternally inherited diseases caused by mitochondrial DNA mutations. The prevalence of hereditary cerebellar ataxia in Europe is estimated to 1–10 per 100 000 (Pratt, 1967; Polo et al., 1991). A multitude of clinical features is associated with autosomal dominant cerebellar ataxia (ADCA) with

both intra- and interfamilial variation (Harding, 1983, 1993).

Narcolepsy is a syndrome of inappropriate sleep attacks, excessive daytime sleepiness, cataplexy, and sometimes hypnagogic hallucinations and sleep paralysis. The diagnosis is clinical and confirmed by multiple sleep latency test (MSLT). The prevalence is 2–7 per 10 000 (Parkes, 1985). The pattern of inheritance of narcolepsy is controversial, proposed to be autosomal dominant (Krabbe and Magnusson, 1942; Daly and Yoss, 1959; Yoss and Daly, 1960; Baraitser and Parkes, 1978) or multifactorial (Bruhova and Roth, 1972; Honda et al., 1984; Guilleminault et al., 1989). In a study of 334 unrelated narcoleptic patients multi-case families with narcolepsy including cataplexy were rare, only 0.9% (Guilleminault et al., 1989). Familial clustering of narcolepsy has previously not been associated with other disease manifestations. Narcolepsy

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Table 1

Approximate ages at onset of symptoms and detection of signs in ADCA-DN (Autosomal Dominant Cerebellar Ataxia Deafness and Narcolepsy). EDS: excessive daytime sleepiness; ?: not known; Yes: present; No: absent. Empty spaces: not relevant.

Patient	II.6 female	III.1 male	III.2 male	III.3 female	IV.4 male
Debut dyscoordination	40–44	28	10	early	child
Ataxia detected	44	38	37	35	12
Debut hearing deficit	38	37	32	35	4
Bilateral total deafness	49	47	No ^a	41	No ^b
Pale optic disks detected	44	38	43	40	No
EDS and sleep attacks	30	child	18	34	?
Cataplexy	39	22	32	34	No
Sleep paralysis	?	26	Yes	No	?
Hypnagogic hallucinations	?	Yes	Yes	No	?
REM sleep behavior disorder	?	Yes	Yes	?	?
Pathological slow pursuit	?	?	42	41	14
Nystagmus	No	No	39	No	No
Bilateral cataracts	No	No	46	No	No
Tremor	?	28	39	41	No
Writer's cramp	No	28	Yes	No	No
Pseudobulbar signs	No	Yes	Yes	Yes	No
Impotence		45	44		
Excessive sweating	?	Yes	Yes	Yes	No
Psychosis	?	44	48	No	No
Epilepsy	No	No	48	No	No
Diabetes mellitus	45	No	47	No	No
Cardiomyopathy	No	46	No	No	No
Life span	51	49			

^a Flat loss hearing deficit of 80 dB bilaterally at age 48.

^b Sloping hearing loss for tones higher than 500 Hz of the left ear age 14.

has been associated with familial multiple sclerosis but only one case in each of four families had narcolepsy (Ekbom, 1966). Multiple sclerosis and narcolepsy possibly share a similar genetic susceptibility (Marcadet et al.,

1985; Younger et al., 1991). A linkage to late-onset diabetes mellitus has been reported in Japanese narcoleptic patients (Honda et al., 1986a). Susceptibility to narcolepsy is strongly associated with certain class II HLA antigens

Table 2

Examples of hereditary cerebellar ataxias. Figures are numbers of cases affected. The present family, ADCA-DN, is different from other described hereditary ataxias in associated features. ADCA types I–III refer to Harding's clinical classification (1993). Other ataxias listed here have not been reported with male to male transmission but are included because some features are shared with our patients. A: families with ataxia, deafness, myoclonus and/ or epilepsy (May and White, 1968; Baraitser et al., 1984; Melo and Ferro, 1989) share some of the clinical features with our family. However, ADCA-DN besides narcolepsy also features optic atrophy, pseudobulbar and extrapyramidal signs, psychiatric disease and definite autosomal dominant inheritance with male to male transmission. B: family SAL-306 (Khati et al., 1993). The nature of the visual loss was not described. C: Narcolepsy was reported in a girl who like a brother had spinocerebellar degeneration (Ozawa et al., 1974). Hearing was not mentioned. The parents were unaffected. Narcolepsy present in only one patient in that family could be coincidental. Unlike in our cases ophthalmoplegia was present. Diabetes mellitus and cardiomyopathy occur in Friedreich's ataxia (Harding and Hewer, 1983), a syndrome of autosomal dominant cerebellar ataxia, optic atrophy, neuropathy and pes cavus (Lundberg et al., 1967), mitochondrial disorders and in ADCA-DN.

Cerebellar ataxias	ADCA types			Present family	A	B	C
	I	II	III				
Cerebellar ataxia	+	+	+	5/5	+	7/7	2/2
Ophthalmoplegia	+	±		0/5			2/2
Dementia	+			2/5			
Extrapyramidal signs	+	±		3/5			
Optic atrophy	+			4/5			
Visual deterioration		+		2/5		7/7	
Pigmentary retinopathy		+		0/5			
Deafness				5/5	+	7/7	
Myoclonus				0/5	±		
Epilepsy				1/5	±		
Narcolepsy				4/5			1/2
Cardiomyopathy				1/5			
Diabetes mellitus				2/5			

on chromosome 6p. White narcoleptic patients with cataplexy are HLA DR2⁺ in about 91% (Guilleminault et al., 1989) to 99% (Parkes and Lock, 1989). Presumed monozygotic twins have been discordant or concordant for narcolepsy (Montplaisir and Poirier, 1987; Douglass et al., 1989) and penetrance seems incomplete.

This paper deals with a syndrome in a Swedish family with multi-cases of narcolepsy and concomitant neurological, psychiatric and non-neurological disease. Necropsy findings are described. Due to multi-organ involvement possibly caused by one genetic defect an association to mitochondrial disorder was made. Muscle biopsy was examined for mitochondrial pathology in one patient. Genetic studies focused on the short arm of chromosome 6, including the HLA DR2 locus and the (CAG)_n repeat at the SCA1 locus.

2. Material and methods

We examined 8 individuals (cases III.1-3, IV.1-4, and patient III.2's wife). Information on the diseased individuals II.1, II.6 and II.8 is from hospital records and relatives, and for the other individuals in generation I and II from death certificates and relatives. Ataxia and deafness were present in 5 and narcolepsy in 4 (Tables 1 and 2). We have no indication that other family members were similarly affected (Fig. 1).

2.1. Case reports

2.1.1. Case II.6

This female died aged 51. Hypertension of 230/130 was discovered and treated at age 31. Diabetes mellitus was diagnosed at age 45. She suffered from recurrent urinary tract infections.

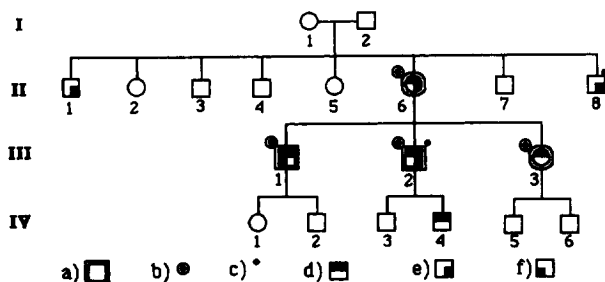


Fig. 1. a: narcolepsy; b: optic atrophy; c: epilepsy; d: ataxia and deafness; e: diabetes mellitus; f: cardiomyopathy. I.1, died aged 85: acute myelogenous leukemia, heart failure and pneumonia. I.2, died aged 66: heart attack. II.1, died aged 82: pneumonia and lung emboli. History of chronic obstructive pulmonary disease, hypertension, late onset diabetes mellitus and strokes. II.2, alive aged 83: presbycusis. II.3, died aged 71: heart attack. II.4, alive aged 76. II.5, alive aged 74. II.6's husband, died aged 75: heart attack. II.7, died aged 58: head injury. II.8, died aged 48: ventricular ulcer hemorrhage. History of chronic alcoholism, pancreas insufficiency, diabetes mellitus and tonic clonic seizures. IV.1-3 and III.2's wife were asymptomatic and normal on examination at ages 27, 16, 16 and 46 years, respectively.

Since the age of 30 or earlier she was tired, frequently fell asleep and sometimes fell. At the age of 39 attacks were noted with twitches in the face. The upper lip pulled upwards and the jaw fell downwards. Breathing was deeper, sometimes snoring, and she would reach for support to keep her balance. Attacks were precipitated by physical or emotional stress, lasted 10–30 s, and occurred one to three times per month. Convulsions were absent. The condition was interpreted as epilepsy at the time and treated with primidone. The symptoms persisted. She was querulant and somewhat paranoid. She was overweight, 88 kg. Visual acuity remained normal. In her forties the optic disks were paler, visual fields moderately constricted and hypertensive retinopathy grade I–II diagnosed. Clinical ataxia was present at age 44 with unsteady broad based gait, limb dysmetria, dyssynergia and Rombergism. Tendon reflexes and sensation including vibration sense were normal. Babinski sign was undecided. Higher cerebral functions deteriorated. She was pronounced legally incompetent at age 51. Terminally she was helpless, completely deaf and unable to walk without support.

2.1.1.1. Autopsy. The brain weighed 1155 g. A slight broadening of the sulci was noticed as well as a slight dilatation of the ventricles. Multiple disseminated softening were present in both white and gray matter of the cerebrum and occasionally in the cerebellum, not in the pons. They varied in size from minute to pea-sized, the largest was found in the basal ganglia of the left hemisphere. The urinary bladder wall was hypertrophic. Ureteroceles were present. The ureters were dilated. Bilateral hydronephrosis and chronic pyelonephritis were diagnosed. The heart appeared normal, but coronary atherosclerosis and a clinically insignificant opening of foramen ovale were described.

2.1.2. Case III.1

This male recalled falling asleep at school. He was tired and fell asleep in monotonous situations but occasionally also when bicycling or eating. Cataplexy, hypnagogic hallucinations and sleep paralysis were present as well as REM sleep behavior disorder. Narcolepsy was treated with methylphenidate for periods. At age 28 his right hand started trembling when writing. He had symptoms of writer's cramp and later preferred typewriting. Hearing deteriorated after age 32. He had a complaint of excessive sweating. Psychiatric symptoms were sudden emotional outbursts. Psychosis developed with paranoia and auditory hallucinations, treated with neuroleptics. At age 38 optic disks were pale. Visual acuity deteriorated from 0.8–1.0 to 0.2 (corrected) in his fifth decade. A systolic heart murmur grade III was noted at age 46. Dependent edema was present and he weighed 111 kg. Hypokinesia, hypomimia, primitive reflexes, and pseudobulbar signs were present. Dysarthria of pseudobulbar type was present. There was gegenhalten of the extremities. The masseter reflex and

deep tendon reflexes were exaggerated but the achilles tendon reflexes were reduced. Babinski sign was present bilaterally. Sensation tested previously was normal. Terminally he was helpless and marasmic, incontinent, completely deaf, fell asleep repeatedly, had poor eye-sight, resting tremor, severe ataxia and was unable to walk without support.

Psychometric test at the age of 44 showed signs of mental reduction. Transaminases were elevated. The ECG showed signs of left ventricular hypertrophy. The relative heart volume was 600 ml/m² radiologically. Echocardiography demonstrated systolic septum thickness 20 mm and left posterior ventricle wall 18 mm. With Doppler technique there was increased supra-aortic flow and slight mitral valve insufficiency. The cardiological evaluation was left ventricular-septum hypertrophy consistent with hypertrophic cardiomyopathy and suspected subaortic narrowing.

2.1.3. Case III.2

This male was mildly unsteady since school age. Ataxia progressed after the age of 38. Balance was markedly worse in the dark. He also had symptoms of writer's cramp. Since age 18 he fell asleep involuntarily, more frequently after his fourth decade. Cataplexy occurred when stressed, excited, or without precipitating factor. Night-mares were frequent. REM sleep behavior disorder, hypnagogic hallucinations, sleep paralysis and automatisms occurred. Narcolepsy was treated with methylphenidate for periods. Hearing deficit and tinnitus were noticed at age 32. Sweating was excessive. Diabetes mellitus was present at age 47. Psychiatric symptoms were difficulty separating dreams from reality, paranoia, visual and auditory hallucinations. He was hospitalized for a period for paranoid psychosis. At age 48 he had repeated generalized tonic-clonic seizures and episodes of partial seizure status epilepticus with disorientation, bizarre behaviour, jerking of the limbs, lip-smacking and contractions around the chin and EEG recording compatible with status epilepticus.

Examinations between ages 37 and 48 revealed progression of ataxia and hearing deficit. Primitive reflexes and pseudobulbar signs developed with dysarthria, exaggerated masseter reflex and spastic tonus increase of the limbs. The deep tendon reflexes were increased. Babinski sign was present bilaterally. Neck tremor was present. Sensation was normal. Visual acuity deteriorated from normal to 0.5 on the right and 0.4 on the left. At age 46 he had bilateral incipient posterior cortical cataracts. Cataract removal did not improve vision. Optic atrophy with pale papillae and defective color vision were observed. He also had nystagmus. Slow pursuit eye movements were irregular. Saccadic eye movements were normal. Hair pigmentation of body and scalp was uneven with depigmented and normal areas. He was overweight, 93 kg.

Psychometric test at age 47 was normal. Plasma lactate

at rest was elevated once, 3.0 mM, and within the normal twice (0.8–2.0 mM).

2.1.4. Case III.3

This female had brief episodes when the head fell forwards at age 34. She tried to avoid laughing to prevent feeling loose-limbed. She fell asleep in monotonous situations, like watching television. Day-time sleepiness was experienced but was not prominent. Hearing deteriorated after age 35 with repeated episodes of sudden deafness. Sweating was a complaint.

At age 41 her weight was 92 kg. She was completely deaf. Gait was somewhat unsteady as was Romberg's test. Speech was normal. Bilateral limb intention tremor and dysidiadochokinesis of the left arm were noted. Postural tremor of the arms and increased flexor tonus of the left elbow with slight spasticity was present. The masseter reflex and tendon reflexes were exaggerated, more of the left arm. Babinski sign was present on the left. Slow pursuit eye movements were jerky. Saccadic eye movements were normal. Visual acuity was 1.0 bilaterally. Color vision (Tabulae isochromatica 2B) revealed three errors on right, one error on left. The optic disks were pale.

2.1.5. Case IV.4

This male is presently 14 years old. A left sided sensorineural hearing deficit for high tones was discovered at age four. He was clumsy at motor skills and attended special school gymnastic exercises. He snored at night and had apnea periods. By age 12 his parents noted him to be tired early in the evenings and falling asleep easily. Cataplexy was not present. His gait was somewhat unsteady on turning. Speech was normal. Muscle tonus was normal. Tendon reflexes were brisk. Discrete limb intention tremor was present. Plantar responses, sensation, and optic fundi were normal.

2.2. Previous investigations

2.2.1. Audiology

Sensorineural hearing deficit was present in the 5 cases. Case IV.4 had a severe left sided hearing deficit for higher tones at the age of 4. By 14 tonaudiogram was normal on the right. On the left, hearing deficit was 15 dB in the range 125–500 Hz but beyond 500 Hz a steep slope of hearing loss was present and was 80 dB at 2 kHz. The hearing deficit of the four others was diagnosed in their fourth decade. Higher frequencies were lost initially, later also involving lower frequencies. Hearing deficit was bilateral flat loss of 80 dB at age 48 of case III.2. The end stage was bilateral flat loss of 100–110 dB in cases II.6, III.1 and III.3. They were for all purposes totally deaf, unable to hear very loud voices even with a hearing aid, in their fifth decade. Episodes of step-wise deterioration and sudden deafness occurred in cases III.1 and III.3. Discrimination was reduced in the older patients and caloric re-

sponses disappeared, but was normal in case IV.4. Brain-stem auditory evoked potential (BSAEP) gave no response in case III.2 at the age of 39 years. His son, patient IV.4, had bilateral progressive increase in latency of BSAEP.

The EEG was normal in case IV.4. Case II.6 had slow basal rhythm, 6–8 Hz, fluctuating with wakefulness at age 44, and the year later bilateral high voltage slow episodes with right-sided predominance and accentuated by hyperventilation. In cases III.1–3 a continuous generalized EEG abnormality was observed in their fourth decade. Serial EEGs of case III.2 (42–47 years) showed progressive slowing of the basal rhythm, 8–6 Hz. Epileptogenic activity was present later in this case.

Neuroradiology was abnormal in all but one case. Case IV.4, examined with magnetic resonance imaging (MRI) at the age of 14, did not reveal any brain abnormalities. Mild to moderate atrophy of the cerebral and cerebellar hemispheres was found on pneumoencephalography (case II.6), computed tomography (cases III.1–3) and MRI (case III.2). MRI of case III.2 at age 43 showed pathological signal intensities as demonstrated in Fig. 6.

2.2.2. CSF

Cell content, protein concentration and electrophoretic pattern were normal in cases II.6, III.1 and III.2. CSF lactate was normal in case III.2.

2.3. Laboratory investigations

MSLT was performed on cases III.1–3 according to standard procedure (EEG, EOG, EMG) with five 20-min periods at 08, 10, 12, 14 and 16 h. In addition, complete night-time polysomnography (PSG), including respiration, oxymetry and myoclonic monitoring was performed in cases III.1 and III.2.

Muscle biopsies of the right m. vastus lateralis were obtained on two occasions 2 years apart from patient III.2 and examined with light microscopy. 7- μ m thick frozen sections were stained with haematoxylin-eosin, modified Gomori trichrome, PAS, oil red O, myofibrillary ATPase at different pH, NADH-tetrazolium reductase (NADH-TR), succinate dehydrogenase (SDH), and cytochrome *c* oxidase (COX). The second biopsy was examined with electron microscopy (I. Nennesmo).

2.3.1. Biochemical muscle analysis

One part of the second biopsy was used for determination of mitochondrial ATP production and enzyme activities. The control group used for these determinations consisted of 11 healthy subjects (age 33 ± 10 years, mean ± 1 SD), all of whom led a sedentary life without regular physical exercise. Differences from the control group were determined with Student's *t*-test, and $p < 0.001$ was considered significant. The rate of ATP production in mitochondria isolated from fresh muscle was determined as described previously (Wibom and Hultman, 1990). Eight

different substrate combinations were used to test the mitochondrial function: a) pyruvate + palmitoyl-L-carnitine + α -ketoglutarate + malate (PPKM), b) glutamate + malate (Glu + Mal), c) *N,N,N',N'*-tetramethyl-1,4-phenyldiamine + ascorbate (TMPD + Asc), d) α -ketoglutarate, e) palmitoyl-L-carnitine + malate (PalCar + Mal), f) pyruvate + malate (Pyr + Mal), g) succinate + rotenone (Succ + Rot), and h) succinate only (Succ).

The mitochondrial enzyme activities were determined spectrophotometrically at 25°C (GDH at 37°C). In homogenates of fresh muscle, COX and succinate:cytochrome *c* reductase (SCR) were determined (Cooperstein et al., 1950; Cooperstein and Lazarow, 1951), and rotenone sensitive NADH:cytochrome *c* reductase (NCR) as described (Sottocasa et al., 1967). In homogenates of frozen muscle citrate synthase (CS) was determined (Alp et al., 1976) and glutamate dehydrogenase (GDH) was determined as described (Wibom and Hultman, 1990).

Case III.1 died aged 49 of extensive bilateral bronchopneumonia. Tissue samples for histology were taken from the cerebrum (including cortical regions from the four lobes, hippocampus, basal ganglia, thalamus and corpora mamillaria), mesencephalon, pons, medulla oblongata, cerebellum (vermis as well as whole sections of the hemispheres) and different levels of the spinal cord. Sections were stained with haematoxylin-eosin, Luxol fast blue and Bielschowsky (I. Nennesmo). The heart and liver were also examined (C. Grant).

2.4. Genetics

HLA-DR were determined with a PCR based method, using sequence-specific primers identifying the DRB1 polymorphism corresponding to the serologically defined series DR1–DR18 and the presence of DRB3 to DRB5 genes. DNA was amplified using 24 pairs of primers and an internal positive control primer pair (Olerup and Zetterquist, 1992). The primers were purchased from Dynal, Oslo, Norway. PCR products were visualised by electrophoresis in ethidium bromide-stained 2% agarose gels. Two patients (III.1 and III.2) were only analysed with standard microcytotoxic method using separated T cells as targets (Vartdal et al., 1986). The serological method could not discriminate between the two splits DR15 and DR16 of DR2.

The size and segregation of (CAG)*n* alleles at the spinocerebellar ataxia type 1 (SCA1) locus at chromosome 6p22 was analyzed by PCR essentially as described (Orr et al., 1993). Fifty ng of genomic DNA was mixed with 10 pmol of each primer (for 5'-GAACTGGAAATGTG-GACGTAC and rev 5'-CCAACATGGGCAGTCTGAG) in a total volume of 25 μ l containing 1.5 mM MgCl₂, 250 μ M dNTPs, 50 mM KCl, 10 mM Tris-HCl pH 8.3 and 1 U AmpliTaq (Perkin Elmer). The forward primer was labelled at the 5' end with [³²P]dATP. Samples were denatured at 94°C for 5 min, followed by 30 cycles of

denaturation (94° C, 1 min), annealing (55° C, 1 min) and extension (72° C, 2 min). 6 μ l of the PCR reaction was mixed with 4 μ l formamide buffer, denatured at 80° C for three min, and electrophoresed through a 6% denaturing polyacrylamide gel (Fig. 5).

The chromosome analyses were made on lymphocytes stimulated with phytohemagglutinin of patient III.2. Ordinary G-banding and high resolution G-banding were used. Twenty cells were karyotyped.

3. Results

3.1. MSLT and PSG

Cases III.1 and III.2 had three REM sleep periods and case III.3 had two. The average REM latency was 13, 11 and 12 min in patients III.1, III.2, and III.3, respectively. These results together with their symptoms fulfill the criteria for diagnosis of narcolepsy as two or more REM periods were observed on the MSLT (Baker et al., 1986). Night-time PSG in patients III.1 and III.2 did not show significant apneas.

3.2. Muscle morphology and biochemistry (case III.2)

Muscle biopsies from m. vastus lateralis were taken at 46 and 48 years. In the first an increase in lipid content of

the muscle fibers appeared to be present. Reactions for NADH-TR and SDH were ordinary but an occasional fiber seemed to be negative for COX. No ragged-red fibers were found. In the second biopsy only a few fibers with a slight increase in lipid content were noticed. Electron microscopy of the second biopsy revealed no obvious changes. The mitochondria had a normal appearance without large accumulation.

The mitochondrial ATP production in muscle was decreased for the substrates PPKM, Glu + Mal, TMPD + Asc and α -ketoglutarate (Fig. 4). This decrease could not be located to any specific site in the mitochondria. The enzymes in the electron transport chain showed normal activities, and were for COX 10.5 (9.6 ± 1.8), SCR 2.6 (2.8 ± 1.0) and NCR 2.4 (2.9 ± 0.6). All enzyme activities are given in $\text{mmol} \cdot \text{min}^{-1} \cdot \text{kg}^{-1}$ muscle (control value ± 1 SD). In the matrix the activity for GDH was decreased, 0.84 (1.33 ± 0.20), while CS was normal, 15.5 (20.0 ± 3.7).

3.3. Autopsy (case III.1)

The formal-fixed brain weighed 1400 g, and the cerebellum 155 g. Macroscopically no specific changes were noted, no obvious signs of atrophy of the cerebellum were present. Microscopical investigation of different regions of the cerebral cortex showed a normal cytoarchitecture. No neuritic plaques or neurofibrillary tangles could be found

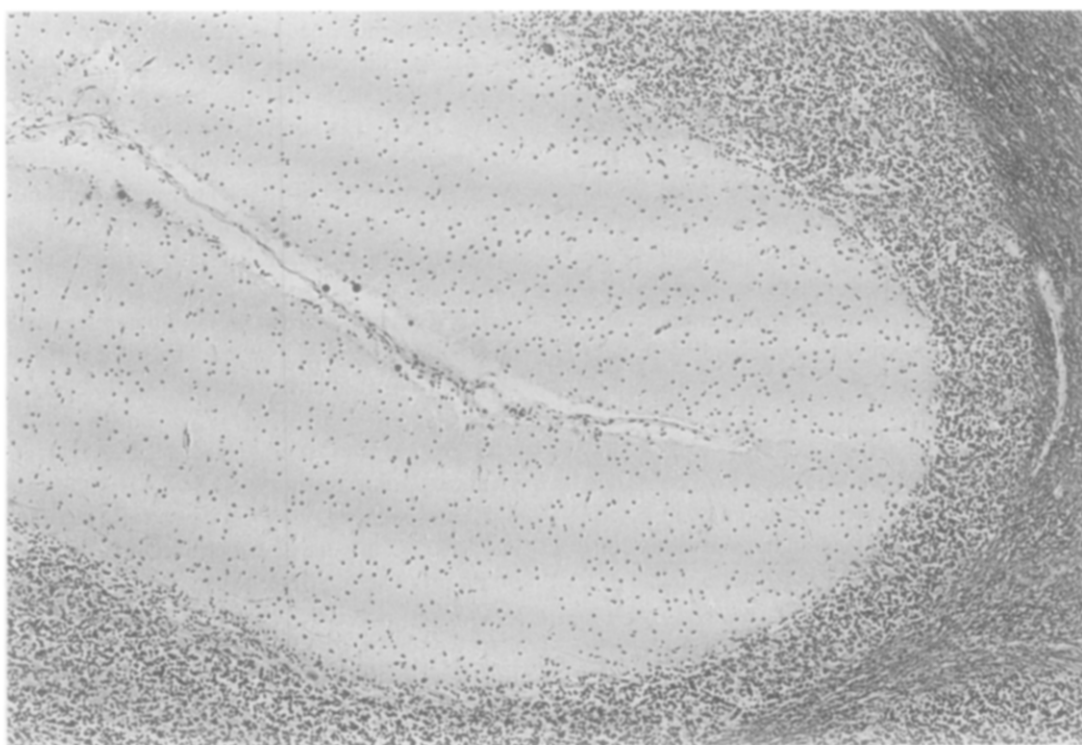


Fig. 2. Section of the cerebellum. A prominent loss of Purkinje cells is seen as well as gliosis of the molecular layer. Luxol fast blue, $\times 93.5$.

in silver stained sections. A few neuritic plaques were present in the hippocampus. In the globus pallidus calcification of vessel walls was seen. The thalamus and corpora mamillaria were without obvious changes. An ordinary number of pigment containing neurons seemed to be present in the substantia nigra. The nuclei as well as transverse fibers of the pons had a normal appearance. A pronounced loss of nerve cells was found in the inferior olives in sections of the medulla oblongata, with gliosis. A variable slight to moderate loss of Purkinje cells was seen in the cerebellar hemispheres as well as in vermis (Fig. 2). In some regions no Purkinje cells were present for a long distance. An increase of glial cells in the molecular layer was seen while the granular cell layer appeared intact. A slight reduction of neurons in the dentate nucleus was

noticed with increased number of glial cells. In the spinal cord the most prominent finding was degeneration in the posterior column of especially the fasciculus gracilis (Fig. 3). Some pallor of spinocerebellar tracts seemed also to be present. In the white matter of the cerebrum mostly, but also in other parts of the brain, thickened vessels with hyalin changes were found. Rather many corpora amylacea were also detected in the white matter of the cerebrum as well as vermis and in the periphery of the spinal cord.

The heart weighed 390 g, slightly more than normal. Left ventricular hypertrophy (12–14 mm) resulted in a conical form with septal bulging into the outflow tract with linear subendocardial fibroelastosis. Histologically, hypertrophied myocardial fibrosis was most prominent in the subendocardial region throughout the left ventricle. In

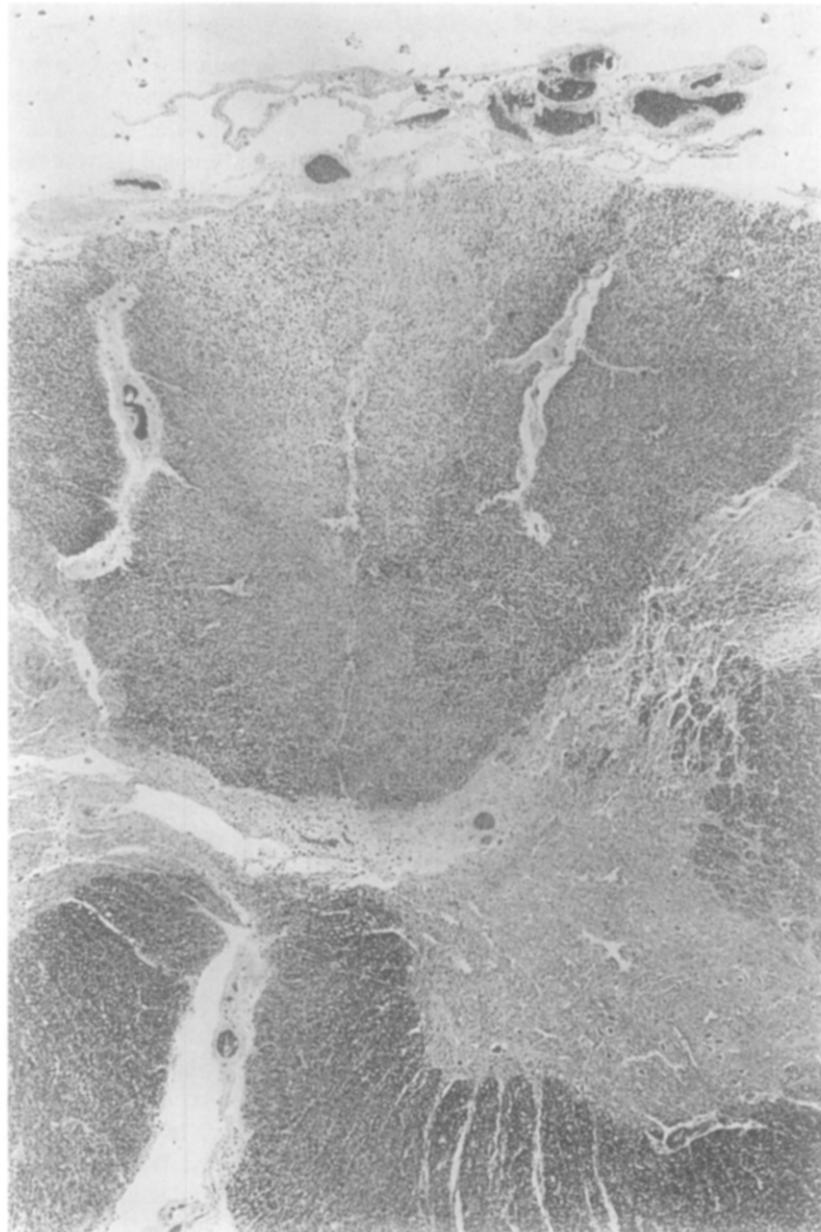


Fig. 3. Transverse section of the spinal cord. Myelin stain showing pallor of posterior columns, especially the fasciculus gracilis. Luxol fast blue, $\times 22.5$.

other respects the heart was within normal for the age. The liver was somewhat enlarged and congested and weighed 1650 g.

3.4. Genetic findings

3.4.1. HLA DR2

Patients III.1-3 were HLA DR2⁺, whereas IV.1-2, 4 and the mother of the latter were HLA DR⁻.

The PCR amplified fragments from the (CAG)_n repeat at the SCA1 locus revealed that the four affected family members tested carried alleles in the normal size range (28–31 repeat units). The segregation of alleles could be followed and linkage to this locus was excluded (Fig. 5).

The chromosome analyses showed an apparently normal male karyotype.

4. Discussion

A hereditary symptom constellation such as in this family has not been reported earlier. The simultaneous presence of ataxia, deafness and narcolepsy in four individuals in the same family is unlikely to be coincidental. In these four, cases II.6 and III.1-3, the disorder was rather uniform with symptoms of narcolepsy and marked progress of ataxia and hearing deficit in the fourth and fifth decades. The diseased II.6 had narcolepsy clinically based on description of excessive daytime sleepiness, sleep attacks and cataplexy. Her three children with similar symptoms had narcolepsy verified by MSLT. The fifth patient (IV.4) had ataxia and hearing deficit. A stable period of several years of mild symptoms of dyscoordination was present in two (cases III.2-3) before progression. Hearing deficit and ataxia were much earlier in case IV.4 compared to his paternal grandmother. Mild symptoms of ataxia and dyscoordination appeared before the age of 20 in 3 patients which is unusual for ADCA syndromes (Harding, 1983, 1993). Age at onset of narcolepsy also showed a wide range.

With increasing disease duration optic atrophy developed in 4 patients and vision deteriorated in 2 of them. Other features were pseudobulbar and extrapyramidal signs, postural tremor, irregular slow pursuit eye movements and prominent sweating in 3 of the patients. Writer's cramp, mental deterioration, psychosis, impotence, and disturbed sphincter control occurred in two. Case III.2 had nystagmus. He also had bilateral cataracts. Generalized tonic clonic seizures and partial seizure status epilepticus also developed in this patient. Two patients became helpless and died prematurely.

The sensorineural hearing loss in this family is comparable to that described in a large pedigree with maternally inherited multi-system mitochondrial disease (Elverland and Torbergson, 1991). Age at onset of hearing deficit varied from 14 to 50 years in that family versus 4 to 38

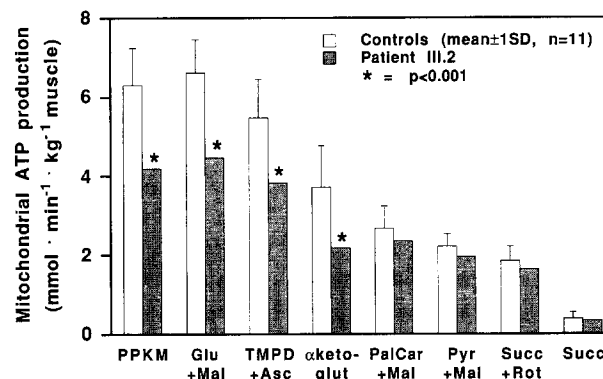


Fig. 4. ATP production in isolated muscle mitochondria in the presence of different substrates in a control group and patient III.2. Significant differences from the control group are indicated (Student's *t*-test, *p* < 0.001). For abbreviations see Material and methods.

years in our patients. The hearing loss in both families first involved higher frequencies, later included lower frequencies and finally flat loss. The rate of hearing loss in different family members was fairly constant in the maternally inherited disease, and only one patient out of 26 in that family was known to have an episode of sudden deafness in one ear. This contrasts to a more unpredictable and faster course of hearing loss in our patients with repeated episodic deterioration and sudden deafness occurring bilaterally in two out of five patients.

The combination of ataxia, epilepsy, deafness, optic atrophy, cardiomyopathy and diabetes mellitus gives association to mitochondrial disorder. The biochemical investigation in muscle of patient III.2 indicated a mitochondrial dysfunction. The mitochondrial ATP production was decreased for the substrates that normally give the highest rates, i.e. PPKM, Glu + Mal, TMPD + Asc and α-ketoglutarate (Fig. 4). This decrease was not associated with any specific malfunction in the electron transport chain, and the activities of the enzymes in the electron transport chain were all normal. The activity for the matrix enzyme GDH was low. This indicates a low mitochondrial density, as has been shown previously (Wibom and Hultman, 1990). The findings in patient III.2 could be an effect of physical inactivity, but this normally renders a general lowering in mitochondrial ATP production for all substrates, rather than the selective decrease found in the patient.

Although a specific enzyme deficiency could not be established, it should be pointed out that mitochondrial proteins are encoded by both nuclear DNA and mitochondrial DNA, thus both Mendelian and maternal inheritance is possible (DiMauro et al., 1992). However, this syndrome is inherited in an autosomal dominant fashion with high penetrance. Male to male transmission excludes X-linked inheritance. Transmission from the male (case III.2) to one of his children excludes a mitochondrial DNA mutation as mitochondrial DNA is exclusively maternally transmitted.

Ataxia, deafness and narcolepsy appeared in any order, the onset of individual symptoms sometimes separated by several years. This does not exclude a single causative genetic factor as a large range of intrafamilial variation of manifestations often occurs in autosomal dominant diseases.

A genetic link between cerebellar ataxia and narcolepsy has hitherto not been described. One possibility is deletions of contiguous genes at closely linked loci, one for narcolepsy and one for ataxia and hearing deficit. Cytogenetic prophase banding was normal and the combined results from HLA typing and analysis of the trinucleotide-repeat at the SCA1 locus exclude linkage to a 15 cM (centiMorgan) interval of chromosome 6 (Figs. 5 and 6). The genetic factor therefore must be located outside this region of chromosome 6. Furthermore, linkage to the sex chromosomes or mitochondrial DNA was excluded due to the pattern of inheritance. The absence of narcolepsy in case IV.4 who was affected by hearing deficit and ataxia is unclear. We cannot exclude that he, still young, develops narcolepsy later in life. A more likely explanation is that the mutant allele that predisposes for narcolepsy needs the presence of another inherited factor, such as HLA DR2, for its expression. This would be in agreement with a proposed multifactorial mode of inheritance of narcolepsy (Bruhova and Roth, 1972; Honda et al., 1984; Guilleminault et al., 1989). The HLA DR2 haplotype is, however, neither sufficient nor necessary for development of narcolepsy (Guilleminault et al., 1989). Incomplete penetrance

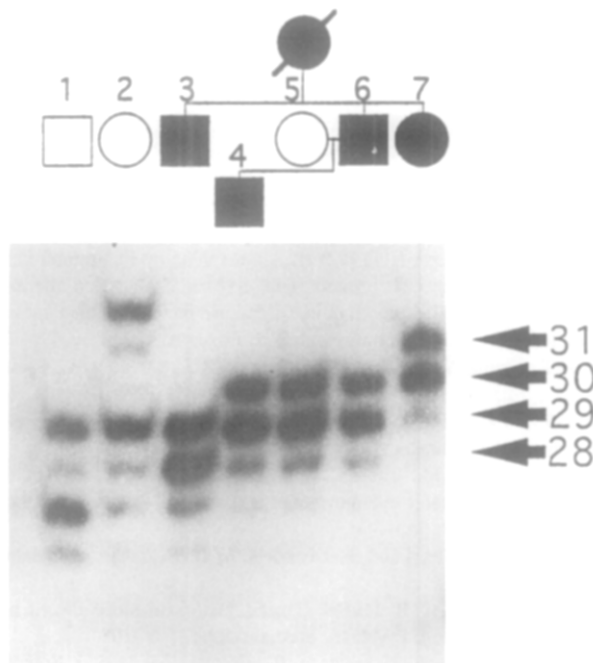


Fig. 5. Analysis of PCR amplified products containing the trinucleotide repeat. Lanes 1 and 2: normal male and female, respectively. Lanes 3–7: members of the family presented. The pedigree is a selected part of Fig. 1. Dark symbols: affected individuals. The number of (CAG)_n repeat units is indicated to the right.

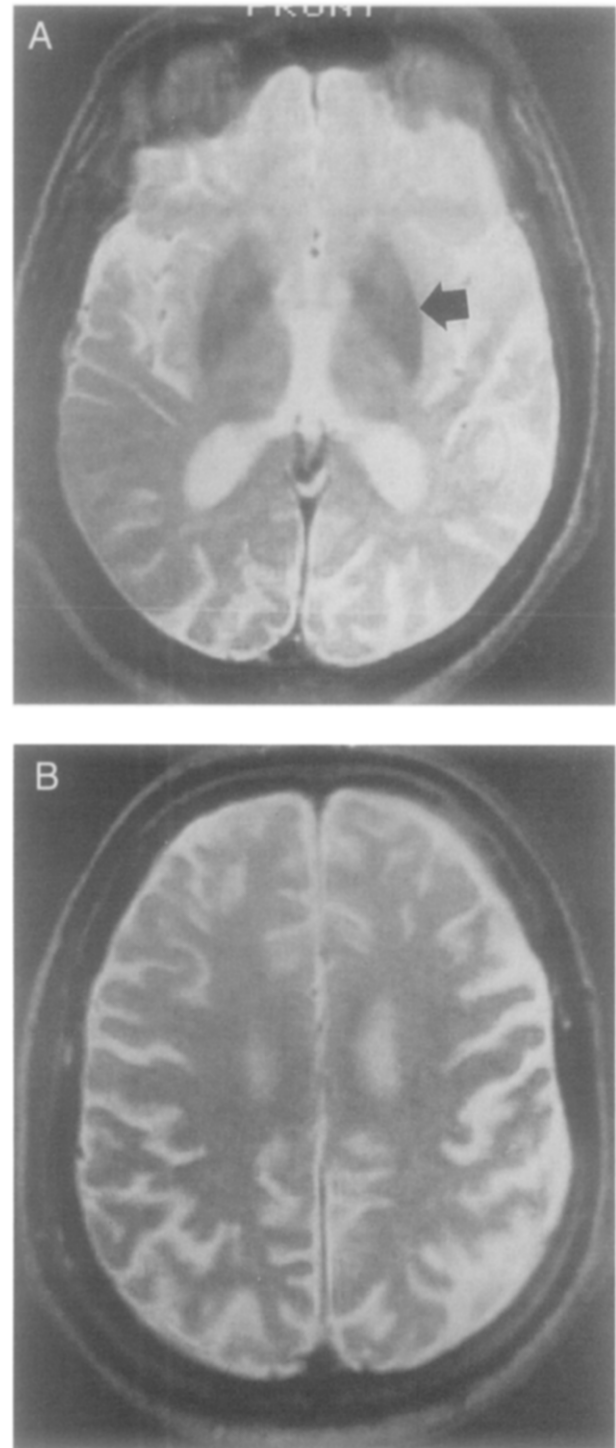


Fig. 6. a–b: MRI of case III.2 at age 43. The normal signal intensity difference between white and gray matter has disappeared. The putamina (arrow) show abnormal low and inhomogeneous signal intensity (a). Enlargement of the third ventricle (a) and cerebral sulci.

of narcolepsy may also account for the absence of narcolepsy in his case.

Phenotype may in some instances be a result of an interaction between one particular mutation and the genetic environment in which it occurs. This may be the situation regarding a mutant allele for narcolepsy expressed in the

presence of a specific HLA type. As only 0.2% of HLA DR2⁺ individuals develop narcolepsy (Parkes and Lock, 1989) the reason might be absence of an additional necessary mutant allele for narcolepsy.

The role of HLA DR2 is unknown in narcolepsy. No immunologic basis for narcolepsy has been substantiated (Honda et al., 1986b; Rubin et al., 1988; Strohmaier et al., 1988). Our preferred interpretation is that the syndrome in this family is the result of a nuclear genetic factor affecting mitochondrial function and influenced by the HLA system.

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